

NITK-UsoundSim

Project flow: which team's output became which team's input

Modular, physics-based 2-D ultrasound simulator | Philips L12-4 reference model: 128-element linear array, 8 MHz | phantom in, B-mode image out

What this document shows

- **Page 2 - the data-flow diagram.** One box per pipeline stage, coloured by team. Each arrow is labelled with the data handed to the next stage: its name, array shape and units.
- **Page 3 - the handover table.** For every team: what it receives and from whom, what it does, what it produces, who uses it next, and which file does it.
- **Page 4 - how the project was built,** step by step, with the check that verified each step.
- **Page 5 - final results** for the three required phantoms (point targets, speckle, cyst).
- **Page 6 - where we differ from the plan,** and why.

The pipeline in one line

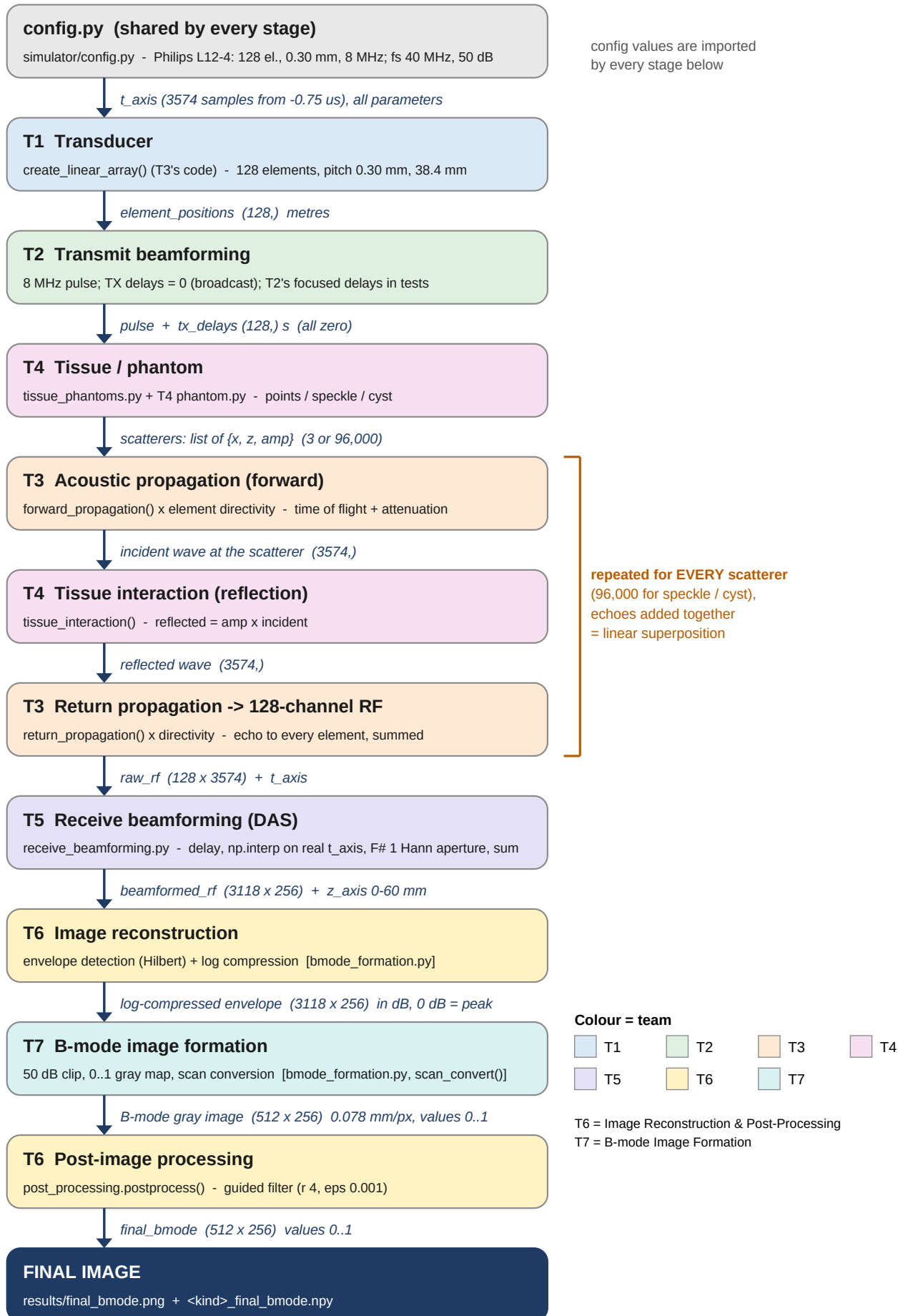
config -> T1 element positions -> T2 transmit pulse and delays -> T4 phantom -> [for every scatterer: T3 forward propagation -> T4 reflection -> T3 return propagation] -> T4 128-channel RF -> T5 delay-and-sum beamforming -> T6 image reconstruction (envelope detection + log compression) -> T7 B-mode gray image formation -> T6 post-image processing (despeckling) -> final B-mode image.

Run it

`python3 simulator/main.py` (all three phantoms, about 8 min on 8 cores) - or open `demo.ipynb` and Run All to see every team's output stage by stage.

Integration and T6 (Image Reconstruction & Post-Processing): Darshil Maniya (262SP009) | 26 September 2026

Data flow between teams



Team-by-team handover

Shapes are for the default run (Philips L12-4 model): 128 elements, 3574 RF time samples (40 MHz), 256 scan lines, 3118 depth samples (0-60 mm), 512 x 256 display pixels.

Team	Receives (from)	Does	Produces (name, shape, units)	Goes to	Code
config	-	One shared set of parameters, imported from T3's config plus integration settings; real RF time axis.	C, FS, DYNAMIC_RANGE_DB from T3; probe N_ELEMENTS 128, PITCH 0.30 mm, F0 8 MHz; t_axis (3574,) s from -0.75 us	every stage	simulator/config.py
T1 Transducer	N_ELEMENTS, PITCH (config)	Places the 128 elements of the Philips L12-4 array (T1's own code is a piezo-material registry, so T3's geometry function is used).	element_positions (128,) m; x = lateral, all at z = 0	T2, T3, T5	T3: create_linear_array(), via config.py
T2 Transmit beamforming	element_positions (T1)	Transmit pulse and per-element transmit delays. Imaging uses a broadcast transmit (all zero delays); T2's focused delays are used in the parameter tests and demo.	pulse (8 MHz, 3-cycle Gaussian); tx_delays (128,) s = 0; transmit field (demo)	T3	T3: generate_pulse() config: get_tx_delays() T2: beamformer.py
T4 Phantom	phantom settings (config)	Builds the object to image: 3 point targets, or 96,000 random scatterers (speckle), or the same with a 6 mm anechoic cyst.	scatterers: list of {x, z, amp} (m, m, a.u.)	T3	T4 folder: tissue_phantoms.py (build_phantom) + phantom.py
T3 Propagation (forward)	element_positions, tx_delays, pulse (T1, T2); one scatterer (T4)	Sum of the pulses from all 128 elements arriving at the scatterer: delay d/c, attenuation 0.5 dB/MHz/cm, each weighted by the element directivity.	incident wave at the scatterer (3574,); travel times (128,) s	T4	T3: forward_propagation()
T4 Tissue interaction	incident wave (T3)	Reflection: scales the wave by the scatterer's strength (amp x (1+ R) x scale).	reflected wave (3574,)	T3	T4: tissue_interaction()
T3 Propagation (return)	reflected wave (T4)	Sends the echo back to all 128 elements (delay + attenuation, x directivity). Repeated for every scatterer; the echoes are added.	echoes (128, 3574) per scatterer, summed	T4	T3: return_propagation()
T4 RF generation	summed echoes	The 128-channel RF data a real probe would record (simulated in parallel on all CPU cores).	raw_rf (128, 3574) + t_axis (3574,) s	T5	T4 folder: tissue_phantoms.py (simulate_rf)
T5 Receive beamforming	raw_rf, t_axis (T4); element_positions (T1); scan lines (config)	For every scan line and depth: round-trip delay to each element, read with np.interp on the REAL t_axis, Hann window over an F# = 1 dynamic aperture, sum (dynamic receive focus).	beamformed_rf (3118, 256) + z_axis (3118,) m, 0-60 mm	T6	T5 folder: receive_beamforming.py (das_beamform)
T6 Image reconstruction	beamformed_rf (T5)	Envelope detection: Hilbert transform along depth, [analytic signal]. Log compression: 20 log10(envelope / max).	log-compressed envelope (3118, 256) in dB (0 dB = peak)	T7	bmode_formation.py (Hilbert + 20 log10 steps)
T7 B-mode image formation	log-compressed envelope (T6); scan lines, z_axis	Clip to the 50 dB dynamic range and map to 0..1 gray levels; scan conversion for a linear array: straight rectangular resample (no fan), 0-40 mm, square pixels.	B-mode gray image (512, 256), 0.078 mm/px, values 0..1	T6	bmode_formation.py (clip + 0..1 map); post_processing.py (scan_convert)
T6 Post-image processing	B-mode gray image (T7)	Guided filter chosen in the Post-Image Processing notebook (10 filters compared on breast, thyroid, kidney, liver images); r = 4, eps = 0.001.	final_bmode (512, 256), values 0..1	results/	post_processing.py (postprocess)

Team numbers follow the team's own split: T6 = Image Reconstruction & Post-Processing, T7 = B-mode Image Formation (plan and folders use the earlier numbering). **Files:** integration in simulator/; each teams/ folder holds the team's original code, unchanged, plus its stage module written during integration.

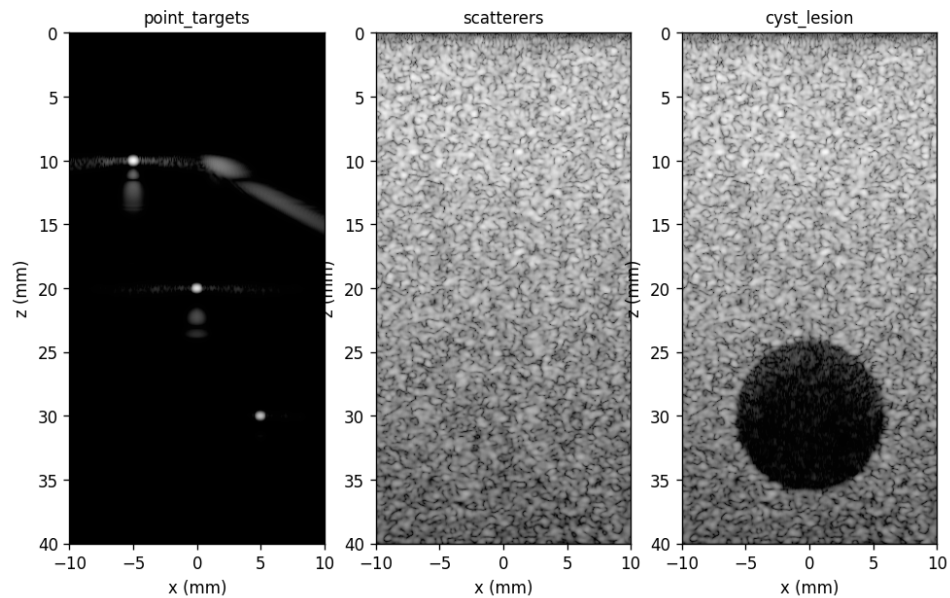
How the project was built

Steps 1-11 built and verified the pipeline with T3's first array (64 elements, 5 MHz, $\lambda/2$ pitch); step 12 moved it to the Philips L12-4 model and re-verified everything.

#	Step	What was done	How it was checked
1	Audit every team's delivery	Read and ran each team's code to see what really exists.	T3 22/22 tests pass, T4 13/13, T2 2/2. T1: no array geometry. T5: test file only. T6: notebook on real carotid data.
2	One shared config	config.py imports T3's constants instead of copying them; defines the real RF time axis (starts at -1.175 us) and the rule: every delay lookup uses np.interp on that axis.	Used by every module; a wrong tau*FS lookup would put targets ~0.9 mm too deep.
3	RF generation (T3 + T4)	Called T3 forward -> T4 reflection -> T3 return for every scatterer, one broadcast transmit.	T4 inside T3's chain gives output bit-identical to T3's own model.
4	Receive beamforming (T5)	Built das_beamform using T5's test file as the specification.	T5's own tests: 4/4 pass (depth errors 0.02-0.05 mm).
5	First point-target image	Checked target positions; found an edge-wave ghost near the array edge.	Targets within 0.08 mm; ghost measured at -25 dB.
6	Image reconstruction (T6) + B-mode gray image (T7)	Envelope detection, log compression and 0..1 gray mapping: the B-mode team's carotid-notebook algorithm, rewritten as bmode_formation().	Matches the B-mode team's own saved carotid output to 5e-6.
7	Speckle and cyst images; ghost fix	Rendered 16,000-scatterer phantoms. The ghost was strong outside the aperture, so the image was limited to +/-5 mm (TX apodization was tested and rejected).	Speckle statistics 1.79-1.90 (theory 1.91); edge-wave level -24 to -32 dB inside +/-5 mm.
8	Scan conversion (T7) + post-image processing (T6)	Rectangular scan conversion for the gray image; guided filter taken unchanged from the Post-Image Processing notebook.	Filter output bit-identical to the notebook's function.
9	One command: main.py	Runs all three phantoms end to end and saves images and arrays.	About 95 s; two runs give bit-identical results.
10	Parameter tests (plan, 24 Sep)	Changed frequency, focus, attenuation, apodization, sampling rate.	9/9 checks pass; e.g. attenuation -9.99 dB vs -10.00 predicted.
11	Documentation and demo	README, MODULES.md (8 items per module), demo.ipynb (team by team), project folders arranged per team.	Notebook runs with 0 errors.
12	Probe update: Philips L12-4	128 elements, 0.30 mm pitch, 8 MHz; image +/-10 mm with 256 lines; element directivity added (coarse pitch gave -11 dB ghosts); $F\# = 1$ receive aperture (grating-lobe streak); 96,000 scatterers simulated in parallel.	T5 4/4, parameter tests 9/9; speckle 1.73-1.92; cyst gCNR 0.86; ghosts -21 to -34 dB, streak below -100 dB.

Final results (python3 simulator/main.py, 50 dB)

NITK-UsoundSim final images (50 dB, guided despeckle)



Phantom	What is checked	Result
Point targets	Position of (-5, 10), (0, 20), (+5, 30) mm; resolution	Found at (-4.98, 10.01), (-0.04, 20.00), (+4.98, 29.99) mm; axial 0.33-0.35 mm, lateral 0.47 mm
Scatterers (speckle)	Speckle statistics (mean/std of envelope; fully developed speckle = 1.91)	1.73-1.92 from 6 to 38 mm; uniform across +/-10 mm; speckle index 0.178 -> 0.163 after the filter
Cyst (6 mm, anechoic)	Contrast against the surrounding tissue	-19.4 dB; CNR 2.323 -> 2.425, gCNR 0.860 -> 0.871 after the filter; edges kept 96 %
Parameter tests	Frequency, TX focus, attenuation, apodization, sampling rate	9/9 checks pass

Where we differ from the plan, and why

Topic	What we did	Why
Probe	Philips L12-4 reference model: 128 elements, 0.30 mm pitch, 8 MHz (plan: 64 elements).	Team decision to model the L12-4. Philips publishes 128 elements and 4-12 MHz; pitch and centre frequency are T2's documented assumptions.
Transducer geometry	From T3's <code>create_linear_array()</code> , not T1's code.	T1 delivered a piezo-material registry, not element positions.
Transmit	One broadcast transmit (all 128 elements at once), no transmit focusing in the images.	Per-line focusing needs one full simulation per line (256 lines x about 4 min). T2's focusing works (+18.4 dB at the focus in the parameter tests).
Pulse	T3's 3-cycle Gaussian pulse (plan: 2-cycle).	Kept T3's tested pulse.
Element model	Element directivity $\text{sinc}(w \sin(\theta)/\lambda)$ added on transmit and receive (T3's elements are points).	With the 0.30 mm pitch (1.56 λ) point elements gave ghosts only -11 to -13 dB below each target; now -21 to -34 dB.
Receive aperture	Dynamic aperture, $F\# = 1$.	Removes a grating-lobe streak (-17 dB -> below -100 dB); lateral width 0.31 -> 0.47 mm.
Cyst	6 mm radius (T4's default), anechoic (a fluid cyst).	Clearly visible: -19.4 dB, gCNR 0.86.
Body model	Point-scatterer phantoms (Field II-style): uniform tissue, one sound speed, straight rays.	A real body model needs CT/MRI tissue maps and a full-wave simulator; the plan asked for point, speckle and cyst phantoms first.

Full details: docs/README.md (decisions and test history) and docs/MODULES.md (explanation, inputs, outputs, equations, algorithm, assumptions, limitations and test result for every module).